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Acoustic-Driven Generation of Pathological Speech Reports Using Large Language Models

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ABSTRACT

Clinical reports compile patients’ histories, treatments, and outcomes, enabling the creation of personalized and effective treatment plans. However, speech disorders are rarely analyzed using such reports, primarily due to the absence of standardized speech protocols. Nevertheless, speech and language therapists (SLTs) can rely on perceptual evaluations, such as the modified Frenchay Dysarthria Assessment (mFDA) scale, to quantify the severity of symptoms across seven categories: breathing, lips, larynx, palate, monotonicity, tongue, and intelligibility. In this paper, we propose using Large Language Models (LLMs) to generate FDA-like text reports from audio recordings. Furthermore, we improve *user control* over the input to the LLM by extracting acoustic biomarkers (correlated with the categories from the mFDA) and using them as prompts to the language model. For this, we used speech recordings from 50 Parkinson’s disease (PD) patients and 50 healthy controls (HC), whose audio recordings were assessed by three SLTs according to the mFDA. Structured reports are generated by feeding acoustic biomarkers that are extracted from the speech signals. For this, we only use acoustic biomarkers that are correlated to the seven categories of the mFDA. The results demonstrate that the LLMs can generate reports with a BLEU score of 0.789 for PD and 0.836 for HC, showing the potential of our proposed approach for practical medical applications.

1 Introduction

Human speech holds many insights into our health, and acoustic signals offer a non-invasive analysis method. However, despite the advances powered by artificial intelligence, developing *speech-based* (medical) technology remains challenging. This is the case for clinical conditions such as Parkinson’s disease (PD), a progressive neurological disorder characterized by motor impairments that also affect the speech production process¹. One of the main challenges in diagnosing PD is that there is no standard way to do it. Neurologists rely on clinical history and physical examination to assess patients, and speech is seldom evaluated. PD patients often develop dysarthria, characterized by monotonic pitch, reduced loudness, and imprecise articulation. Although a single modality (speech) is not sufficient to diagnose PD, our previous research has shown that for some patients it is possible to monitor the progression of the speech severity from audio recordings captured over time². Speech is a multidimensional biosignal with a high time—frequency dependent parameter variance³. Acoustic analysis is usually performed for objective assessment by extracting different biomarkers; however, interpreting acoustic information demands expert knowledge in signal processing, reducing the possibility of developing practical medical applications.

Clinical assessment of pathological speech rarely has associated notes or reports, primarily due to the absence of standardized protocols. Speech and language therapists (SLTs) can rely on score-based scales such as the Frenchay Dysarthria Assessment–2 (FDA–2)⁴, which assesses speech production by grading different aspects including swallowing, respiration, lips movement, palate related sounds, laryngeal function (during speech), tongue movement, fluency, and speech intelligibility. Such an evaluation is based on various stimuli, including the sustained phonation of vowels (e.g., producing the vowel /ah/ for three seconds), the rapid repetition of syllables (repeating /pa-ta-ka/ as fast as possible for some seconds), reading isolated words, sentences, and/or text, and having conversations with the interviewer/evaluator. There is also a modified version of the FDA (called mFDA)⁵, which SLTs can administer by listening to the speech recordings, making it suitable for at-home monitoring. The mFDA allows speech evaluation across seven categories: breathing, lip movement, larynx function, palate-produced sounds, monotonicity, tongue movements, and overall intelligibility. While these scoring systems allow clinical assessment of dysarthric speech, the evaluation can be time-consuming, and the quantitative assessment depends on the SLT’s experience, which might limit the evaluation’s accuracy or consistency.

In summary, acoustic analysis is a well-established approach for detecting and monitoring speech symptoms, such as those produced by PD; however, interpretation can be difficult and requires expert knowledge in signal processing. On the other hand, clinical assessment can be carried out using perceptual scales such as the mFDA; which is a subjective and demanding task. To address these challenges, we propose combining large language models (LLMs) with acoustic analysis to generate a clinical report-like text that takes acoustic biomarkers as the primary source of information. In particular, we considered speech recordings of PD patients and healthy controls (HC) assessed by expert phoniatricians according to the mFDA. Combining pathological speech assessment, acoustic analysis, and LLMs can enhance interpretability, efficiency, and accessibility to clinical evaluation by transforming complex acoustic information (e.g., spectral energy) into categorical graduations (from mFDA) of different aspects of spoken language for speech therapists, patients, and healthcare practitioners.

1.1 Related work

Research in automatic medical report generation (aMRG) is not necessarily novel. Early research primarily leveraged limited-size word lexicons within automatic speech recognition systems to dictate radiology reports, representing pioneering steps^{6,7}. Recent advancements have significantly enhanced the capabilities of aMRG. Two primary approaches have emerged in this area: CLIP-based models align modality-text pair embeddings through contrastive learning, enabling efficient cross-modal retrieval and understanding, while multimodal LLMs integrate multiple modalities directly into the language model’s embeddings using modality-specific encoders⁸. The commonly deployed network architecture typically leverages an encoder-decoder structure, where the encoder is used for domain-specific feature extraction and the decoder for the actual report generation. The decoder could be a Transformer variant trained from scratch^{9,10}, or based on a pre-trained LLM. Prominent examples include Vicuna-finetuning^{11–13}, GPT-3¹⁴, OpenFlamingo (which builds on LLaMA and CLIP ViT)¹⁵, or LLaMA¹⁶. Research on aMRG is predominantly applied in several medical fields: the most prominent field is radiology^{10,13,17–19} but other fields include pathology¹⁶, ophthalmology²⁰, and surgical report generation⁹. In all of the methods above, speech has been used as a *tool* to generate reports, e.g., as spoken instructions instead of text. However, to the best of our knowledge, this is the first work of its kind to extend research on aMRG to the pathological speech domain, utilizing patients’ audio signals as the primary source of information to generate the reports.

1.2 Contributions of this paper

This study presents the first approach to generating mFDA-based clinical reports from acoustic biomarkers using LLMs, by extracting features from recordings of PD and HC subjects and tokenizing them as text instructions for the LLM. The reports are based on the mFDA scale, which uses numerical values to quantify the dysarthria severity level. In our approach, we assigned short descriptions (based on the descriptions of the original FDA–2) to each score instead of numerical values to improve interpretation. The mFDA scores were obtained from three phoniatricians who listened and evaluated the audio recordings of PD patients and HCs across four speech tasks: sustained phonation (vowel /a/), repetition of /pa-ta-ka/ (DDK), text reading, and a spontaneous conversation with the interviewer/evaluator. We computed several acoustic biomarkers and selected the most relevant features per mFDA category (breathing, lips, palate, larynx, monotonicity, tongue, and intelligibility) using an statistical approach. The most relevant task-specific features were used to generate input instructions for the LLM. We also propose a data augmentation approach based on the acoustic biomarkers to simulate mFDA clinical reports suitable for fine-tuning the LLMs. Given the limited amount of subjects in our work, we used the simulated reports to help the model learn to interpret the input prompts and also to create the mFDA-based reports following a structured format. To “simulate” the acoustic biomarkers, we use the distribution of the real data. Further details are given in Section 2.5.

In summary, the main contributions of our work are:

- A speech-based approach that takes acoustic biomarkers as prompts to an LLMs that generate mFDA-based text reports, improving interpretability and user controls over the input used by the language model to generate text.
- A data augmentation strategy to simulate acoustic biomarkers and clinical reports. The primary function of such reports is for the model to learn to interpret acoustic-based input prompts and to generate structured speech reports.

Our *main goal* is to show how LLMs can be used to support clinical experts (physicians, neurologist, STLS) to assess patients with speech technology that allows them to keep (up to some extent) control of the input while ensuring interpretability of complex signal processing methods, i.e., acoustic biomarkers.

The *research question* is whether LLMs can effectively learn to associate acoustic biomarkers with clinical speech assessment reports. One key advantage of using handcrafted features instead of neural embeddings is that they provide greater user control over the input, enhancing interpretability and transparency.

The *motivation* behind this work is the almost non-existent approaches in the literature that consider the audio signals as the primary source of information to generate a clinical-like report, instead of using speech as a tool to input spoken prompts to a LLM or to improve the recognition rate on an automatic speech recognition system.

2 Materials & Methods

Figure 1 provides an overview of the proposed methodology for generating mFDA-based reports using LLMs and acoustic biomarkers extracted from audio signals of PD/HC subjects. The participant's speech production was assessed with the mFDA, which quantifies speech on seven categories. We convert these numerical scores into text reports (Section 2.1), extract 248 features from the audios related to phonation, prosody, and phonemic articulation (Section 2.2), select those that correlate with the mFDA (Section 2.3), and transform them into prompts to feed the LLMs (Section 2.5).

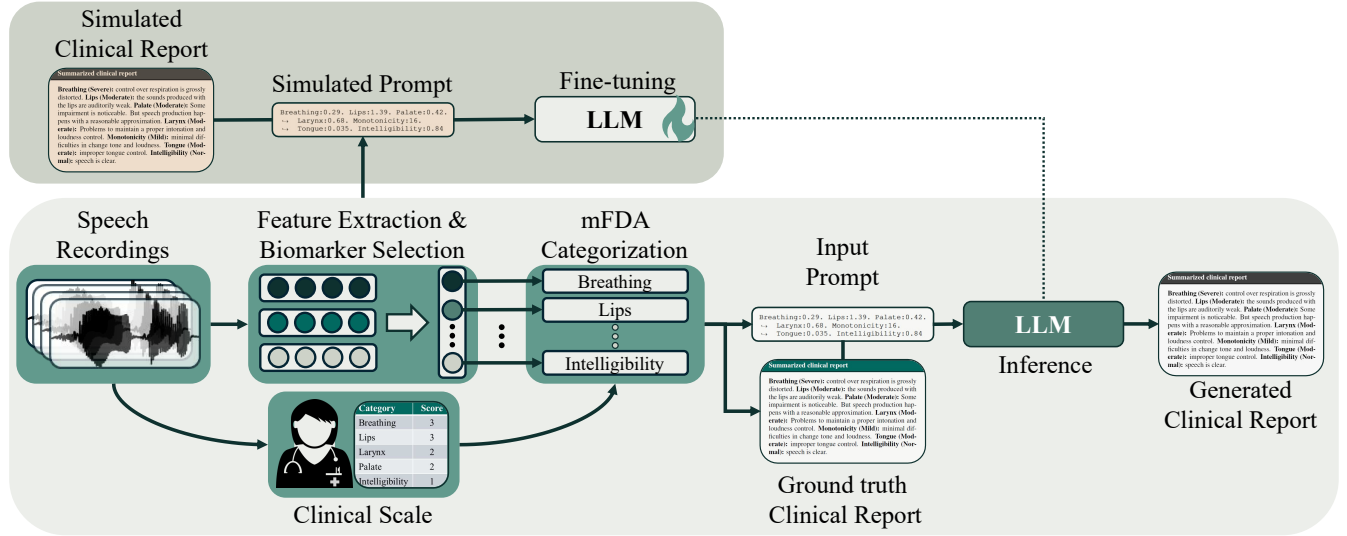


Figure 1. Overview of the proposed methodology for generating clinical reports using LLMs and acoustic biomarkers

2.1 Data

We considered speech recordings of 50 PD patients (25 male / 25 female) and 50 HC (25 male / 25 female) Colombian Spanish native speakers from the PC-GITA dataset²¹. The average age of the PDs is 61.9 ± 9.61 and 60.9 ± 9.46 for HCs. The recordings were captured in a soundproof booth and a neurologist diagnosed all patients. None of the HCs had a neurological disease or presented speech disorders. All audios are resampled at 16kHz. We considered four speech tasks: (1) sustained phonation of vowel /a/, (2) repetition of /pa-ta-ka/, (3) text reading, and (4) spontaneous conversation with the interviewer/evaluator. This study is in compliance with the Helsinki Declaration and was approved by the Ethics Committee of the Clínica Noel, in Medellín, Colombia. A written informed consent was signed by each participant.

mFDA-based clinical reports: Contrary to the FDA-2, the mFDA can be administered by listening to audio recordings⁵ and its one of the few standard scales used to assess pathological speech. The mFDA has seven categories, which an expert phoniatician rates with numbers between “0” (normal) and “4” (severe impairment). In this work, three expert phoniaticians listened to the audios and graded each item of the mFDA with an inter-rater reliability of 0.86. We use the median value of the scores given by the phoniaticians. In this work, we associated the scores to a categorical rating as follows: 0/1: *Normal*, 2: *Mild*, 3: *Moderate*, and 4: *Severe*. Table 1 shows the list of categories and categorical ratings (and associated descriptions) considered in this study. Figure 2 shows the number of subjects (PD/HC) per categorical ratings.

2.2 Acoustic Biomarkers

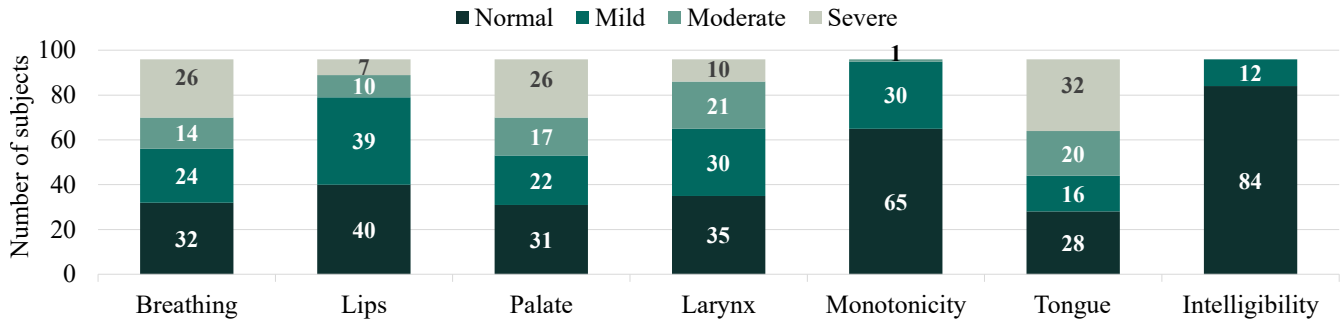
We computed 248 task-specific (Section 2) acoustic biomarkers, proven to be suitable for pathological speech analysis^{3,22–24}. Since the audios were capture in controlled acoustic conditions, no denosing was performed; however, we scaled the amplitude of the audios between -1 and 1 and centered them around zero. The audios were segmented into windows of 20 ms with a step of 10 ms. For features like pitch and loudness, we used windows of 40 ms to include at least three pitch periods. Hanning windowing was applied before feature extraction.

2.2.1 Phonation Features:

Voice disorders are associated with abnormal vibration of the vocal folds and disturbances in respiration that can be detected from from the sustained phonation of vowels (vowel /ah/). Thus, the set of features considered are the standard deviation of pitch ($F0$; to measure vocal fold abnormal vibrations), sound pressure level (SPL; to measure loudness variability) and

Table 1. List of mFDA categories, speech tasks, categorical ratings, and FDA-2-based assessment.

Category	Speech task	Description	Categorical rating with assessment
Breathing	Repetition of /pa-ta-ka/	Fluency breaks	Normal: Normal breathing control Mild: Occasionally breaks fluency due to mild respiratory control Moderate: Requires taking breaks to complete the tasks most of the times Severe: Control over respiration is grossly distorted
Lips	Repetition of /pa-ta-ka/	Lips movement control	Normal: Normal coordination of lips Mild: Inconsistent lip closure, but overall good control Moderate: The sounds produced with the lips are auditorily weak Severe: Unable of maintaining good lip closure
Palate	Repetition of /pa-ta-ka/	Production of velar sounds	Normal: Normal production of velar sounds Mild: Mild abnormality. It does not affect speech production Moderate: Speech production happens with a reasonable approximation Severe: Speech quality is extremely deteriorated
Larynx	Sustained phonation	Pitch/loudness stability	Normal: Can maintain a normal phonation and produce clear speech Mild: Minor difficulties without noticeable speech problems Moderate: Some difficulties to maintain intonation and loudness control Severe: Unable to maintain intonation variations in a conversation
Monotonicity	Spontaneous conversation	Overall prosody	Normal: Can produce variable tones and/or regular pace during speech Mild: Minimal difficulties to change tone and variable pace Moderate: there are problems to vary intonation and/or pace Severe: Speech sounds flat due to limited changes in intonation and pace
Tongue	Repetition of /pa-ta-ka/	Production of /t/	Normal: Tongue movement within normal range Mild: Mild difficulties to move the tongue (not perceptually noticeable) Moderate: Improper tongue control Severe: Severe tongue control
Intelligibility	Reading of text	Overall intelligibility	Normal: Speech is clear Mild: Speech is mostly comprehensible Moderate: Speech is produced with difficulty and it is hard to understand Severe: Speech is incomprehensible

**Figure 2.** Distribution of subjects grouped by category rating. No subjects from PC-GITA had “Moderate” and/or “Severe” ratings in the “Monotonicity” and “Intelligibility” categories

period/amplitude perturbation quotients (PPQ/APQ): PPQ-3 (jitter), PPQ-5, and PPQ-11 to measure short/long term F_0 variations and APQ-3 (shimmer), APQ-5, and APQ-11 to measure short/long term amplitude variations^{25,26}.

2.2.2 Prosody Features:

Speech prosody includes the pitch, loudness, and duration that contributes to production and perception of intonation, rhythm, lexical tone, and stress²⁷. These parameters are computed from the *reading* and *conversational* speech tasks. The features are the pitch and SPL standard deviations (stdF0 and stdSLP) suitable to detect monotonous speech. We also extract the pauses, voiced sounds (speech sounds generated with vibration of vocal folds), and voiceless sounds (speech sounds generated without vibration of vocal folds). The set of features from the extracted segments include the average duration, rate, and number of pause/voiced/voiceless sounds^{23,28}.

2.2.3 Phonemic Features:

These features are computed from the *reading* and *DDK* speech tasks using PhonoQ¹, a deep neural network model that computes pronunciation precision, duration, rate, and rhythm parameters from 18 phonemic groups (including silence) based on voicing (voiced and voiceless phonemes), manner of articulation (stop, nasal, trill, fricative, approximant, lateral, and vowel phonemes) and place of articulation (labial, alveolar, velar, palatal, postalveolar, central, front, and back phonemes)^{24,29}. Pronunciation precision is computed as the average posterior probabilities that a phoneme class (e.g., nasal sounds) is present in the speech recording. Phoneme precision is also measured as the average log-likelihood ratio (LLR). Rhythm-based features are computed using the pairwise variability index (PVI)³⁰. The full set of features is described in the GitHub page of PhonoQ.

2.3 Feature Selection

Statistical analysis was conducted to identify seven acoustic biomarkers associated with the mFDA category ratings. First, an omnibus test of normality study was performed on the acoustic biomarkers. From the results, we learn that not all of the features have a normal distribution; thus, we use the non-parametric we applied Kruskal-Wallis³¹ to all acoustic features considered for analysis by grouping subjects based on their mFDA category ratings. We also computed the point-biserial correlation coefficient (r_{pb})³² and selected the most correlated features to the category ratings. The selected biomarkers with correlation values are shown in Table 2.

Table 2. List of selected biomarkers, associated speech task, correlation with mFDA, and descriptions. The asterisks represent significant correlation with p -values: < 0.05 (*); < 0.01 (**), & < 0.001 (***)

Category	Biomarker	r_{pb}	Speech task	Description
Breathing	Consonants nPVI	0.25*	Rep. /pa-ta-ka/	Consonants duration variability i.e., /p/, /t/, & /k/
Lips	Voiceless LLR	0.24*	Rep. /pa-ta-ka/	Log-precision of voiceless sounds, e.g., /p/
Palate	Velar Max Post	0.20*	Rep. /pa-ta-ka/	Pronunciation precision of velar sounds i.e., /k/
Larynx	APQ-11	0.06	Vowel /a/	Amplitude variability in 11 consecutive glottal cycles
Monotonicity	Stop Avg dur	0.34**	Conversation	Average duration of stop sounds e.g., /p/, /g/, /b/
Tongue	Alveolar dVar	0.29**	Rep. /pa-ta-ka/	Duration variability of alveolar sounds, i.e., /t/
Intelligibility	Voiceless Max Post	0.59***	Text reading	Pronunciation precision of voiceless sounds, e.g., /s/

2.4 Language Models

The mFDA-based clinical reports are generated by fine-tuning several pre-trained LLMs from Hugging Face²: T5 (Text-To-Text Transfer Transformer)³³, Flan-T5 (Fine-tuned Language Net-T5)³⁴, GPT-2 (Generative Pre-trained Transformer)³⁵, LLaMA-7B³⁶, Mistral-7B³⁷, and DeepSeek-R1-Qwen2.5-7B³⁸.

2.4.1 T5 & Flan-T5:

T5 is an encoder-decoder model pre-trained on a mixture of unsupervised and supervised tasks, all formatted as text-to-text tasks. It uses a modified strategy called “span-corruption” based on Masked Language Models. This approach involves replacing contiguous random spans of text with a single mask token. By adding a specific prefix to the input for each task, it performs well across various tasks right from the start. The Flan-T5 model is trained on instruction-tuning tasks, which involve formulating the tasks as explicit instructions to enhance its ability to follow prompts. For fine-tuning, we concatenated the following prefix to the beginning of the input prompt: `Generate a report for: <Instruction>`

2.4.2 GPT-2:

This is a decoder-only autoregressive model, meaning it generates text by predicting the next token. In the GPT-2 model, the inputs are sequences of continuous text of a certain length, and the targets are the same sequence, shifted one token (word or piece of word) to the right. The model uses internally a mask mechanism to make sure the predictions for the token i only use the inputs from 1 to i but not the future tokens. In contrast to T5, GPT-2 does not require prefixes to generate text.

2.4.3 LLaMA-7B:

LLaMA is a collection of open-source foundation models (developed by Meta AI) ranging from 7B to 65B (billion) parameters. The network is based on the transformer architecture with some improvements, such as normalizing the input layers of each transformer sub-layer, using the SwiGLU activation function, removing absolute positional embeddings, and instead adding rotary positional embeddings. The core idea of LLaMA is based on the notion that it is not necessary to scale models to improve

¹<https://github.com/TAriasVergara/PhonoQ/tree/main>

²<https://huggingface.co/models>

performance; instead, it is possible to achieve better results with fewer parameters trained on large amounts of data. In this work, we fine-tuned the LLaMA model with 7B parameters using the following checkpoint: `unsloth/llama-2-7b-bnb-4bit`.

2.4.4 Mistral-7B:

Similar to LLaMA, Mistral was designed to improve LLMs performance without scaling the number of parameters. For this, Mistral leverages grouped-query attention (GQA) and sliding window attention (SWA). GQA significantly accelerates inference speed and reduces memory requirements during decoding, allowing for larger batch sizes. In addition, SWA is designed to handle longer sequences more effectively at a reduced computational cost, thereby alleviating a common limitation in LLMs. These attention mechanisms collectively contribute to the enhanced performance and efficiency of Mistral. We fine-tuned the `unsloth/mistral-7b-bnb-4bit` checkpoint.

2.4.5 DeepSeek-R1-Qwen2.5-7B:

DeepSeek-R1 was designed with the aim of improving reasoning capabilities using reinforcement learning (RL) without any supervised data. DeepSeek-R1 also employs group relative policy optimization³⁹, an RL algorithm that simplifies the process by forgoing a separate "value function" (which estimates future rewards) and instead uses a group-based approach to calculate advantages, leading to reduced memory and computational costs. The checkpoint used in this work, `unsloth/DeepSeek-R1-Distill-Qwen-7B-unsloth-bnb-4bit`, is distilled from DeepSeek-R1 to post-train Qwen2.5-7B Base. For distilled models, DeepSeek doesn't include an RL stage.

2.5 Training

A key challenge we encountered was fine-tuning the LLM without overfitting and having sufficient data to "teach" the model how to structure the reports. For this, we designed an augmentation strategy based on the acoustic biomarkers of the real data to simulate input prompts and their corresponding mFDA reports. We acknowledge that there might be some data leakage in the test set, even though the real data is actually never used for training. However, as shown by the results (Section 3), not all models are able to learn how to structure the reports or correctly associate the acoustic biomarkers to the mFDA categorical rating. In addition, the values used for simulating the prompt won't account for outliers and still would be able to perform relatively well. The procedure is as follows:

1. The PD and HC subjects are grouped based on their categorical ratings on each mFDA category. Then, we compute the mean values for the corresponding acoustic biomarker. Figure 3 shows the average values obtained for each feature. We linearly extrapolated the missing ratings in "Monotonicity" and "Intelligibility" (see the missing ratings in Figure 2).

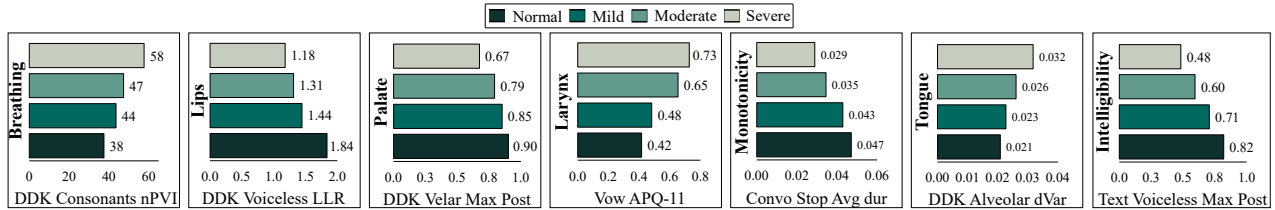


Figure 3. Selected acoustic biomarkers distribution per mFDA category.

2. We set thresholds for the features on each categorical rating by considering around $\pm 5\%$ deviation from the mean values of the acoustic biomarkers. Table 3 shows the respective range of values for the selected features.

Table 3. Range of values used to assign the categorical ratings on the simulated mFDA reports

mFDA Category	Acoustic Biomarker	Categorical rating			
		Normal	Mild	Moderate	Severe
Breathing	DDK Consonants nPVI	$x \leq 40$	$40 < x \leq 45$	$45 < x \leq 50$	$x > 50$
Lips	DDK Voiceless LLR	$x \geq 1.62$	$1.38 \leq x < 1.62$	$1.25 \leq x < 1.38$	$x < 1.25$
Palate	DDK Velar Max Post	$x \geq 0.88$	$0.82 \leq x < 0.88$	$0.73 \leq x < 0.82$	$x < 0.73$
Larynx	Vowel APQ-11	$x \leq 0.42$	$0.42 < x \leq 0.55$	$0.55 < x \leq 0.66$	$x > 0.66$
Monotonicity	Convo Stop Avg dur	$x \geq 0.045$	$0.039 \leq x < 0.045$	$0.032 \leq x < 0.039$	$x < 0.032$
Tongue	DDK Alveolar dVar	$x \leq 0.022$	$0.022 < x \leq 0.025$	$0.025 < x \leq 0.029$	$x > 0.029$
Intelligibility	Text Voiceless Max Post	$x \geq 0.76$	$0.65 \leq x < 0.76$	$0.55 \leq x < 0.65$	$x < 0.55$

- For each category, we generate a random number within the range of valid category values to create the input prompt for the LLMs. For example, an input prompt could look as follows:

```
Breathing: 46 Lips: 1.84 Palate: 0.24 Larynx: 0.27 Monotonicity: 0.057 Tongue: 0.037
→ Intelligibility: 0.39
```

- Finally, the target mFDA report is formed by associating the generated value to the corresponding categorical description (Tables 1 & 3). An example of a mFDA report is shown in Figure 4.

Breathing (Moderate): the subject requires taking breaks to complete the tasks most of the times. **Lips (Normal):** normal coordination of lips. **Palate (Severe):** speech quality is extremely deteriorated. **Larynx (Normal):** can maintain a normal phonation and produce clear speech. **Monotonicity (Normal):** person can produce variable tones and/or regular pace during speech. **Tongue (Severe):** severe tongue control. **Intelligibility (Severe):** speech is incomprehensible.

Figure 4. Example of mFDA-based report for the input prompt: *Breathing: 46 Lips: 1.84 Palate: 0.24 Larynx: 0.27 Monotonicity: 0.057 Tongue: 0.037 Intelligibility: 0.39*

3 Experiments & Results

We randomly generated 110K instructions (input prompts) with associated mFDA-based clinical reports (target output) with the following conditions: 25% of the reports contain random ratings on all categories, 25% contain only *Normal-Mild* ratings, 25% contain only *Mild-Moderate* ratings, and 25% contain only *Moderate-Severe* ratings. The **simulated** mFDA reports are **used to fine-tune** the models by creating a train set (100K reports) and a validation set (10K reports). The **real** mFDA reports (100 reports) from the PD/HC subjects are **used for inference** to measure the performance of the LLMs with the ROUGE, BLEU, and BERT scores⁴⁰. The models were trained on an NVIDIA A100 with 40 GB over three epochs. The learning rate was set to 3e-4, the weight decay to 10e-2, and the batch size was 8. The clinical reports are generated with 512 output tokens, beam search: 3, and no repeating *n*-gram: 2. We fine-tuned three model size variations for T5, Flan-T5, and GPT-2.

3.1 LLM performance on mFDA report generation quality

Table 4 shows the performance of the LLMs evaluated on the 50 PD and 50 HC mFDA reports. Note that we did not fine-tune separate models for PD and HC; instead, we used the same models trained on the simulated mFDA reports and computed the metrics on the real PD/HC reports individually. The table shows that LLMs perform significantly better (two-sided t-test) on

Table 4. LLMs mFDA-based report generation performance on PD and HC groups. **R-1,-2,-L:** ROUGE scores. **AVG:** Overall average score. The best results are highlighted in bold, and the second-best are underlined. Higher is better.

Model	PD reports						HC reports					
	R-1	R-2	R-L	BLEU	BERT	AVG	R-1	R-2	R-L	BLEU	BERT	AVG
T5-60M	0.557	0.280	0.436	0.740	0.901	0.583	0.502	0.241	0.418	0.700	0.890	0.550
T5-220M	0.590	0.331	0.483	0.761	0.907	0.615	0.521	0.273	0.437	0.719	0.896	0.569
T5-770M	0.577	0.306	0.458	0.750	0.905	0.599	0.534	0.275	0.449	0.719	0.895	0.574
Flan-T5-60M	0.554	0.283	0.409	0.732	0.902	0.576	0.479	0.220	0.363	0.695	0.887	0.529
Flan-T5-220M	0.574	0.294	0.432	0.738	0.905	0.589	0.540	0.288	0.442	0.735	0.899	0.581
Flan-T5-770M	0.571	0.275	0.441	0.749	0.901	0.587	0.586	0.336	0.496	0.756	0.900	0.615
GPT-2-124M	0.388	0.206	0.240	0.442	0.854	0.426	0.348	0.167	0.215	0.409	0.847	0.397
GPT-2-355M	0.400	0.211	0.247	0.461	0.856	0.435	0.348	0.164	0.210	0.421	0.848	0.398
GPT-2-774M	0.395	0.204	0.246	0.460	0.855	0.432	0.349	0.166	0.210	0.416	0.847	0.398
LLaMA-7B	<u>0.642</u>	0.441	0.572	0.789	0.937	0.676	0.734	0.599	0.697	0.836	0.950	0.763
Mistral-7B	0.634	<u>0.433</u>	<u>0.568</u>	<u>0.786</u>	<u>0.936</u>	0.672	0.734	<u>0.597</u>	<u>0.695</u>	0.836	0.950	<u>0.762</u>
DeepSeek-7B	0.645	<u>0.433</u>	0.566	<u>0.786</u>	0.937	<u>0.673</u>	<u>0.673</u>	0.491	0.623	<u>0.795</u>	<u>0.939</u>	0.704

HC reports compared to PD reports for LLaMA (PD [AVG]: 0.676 vs. HC [AVG]: 0.763; $p\text{-value} < 0.001$), Mistral (PD [AVG]: 0.672 vs. HC [AVG]: 0.762; $p\text{-value} < 0.001$), DeepSeek (PD [AVG]: 0.673 vs. HC [AVG]: 0.704; $p\text{-value} = 0.173$), and Flan-T5-770M (PD [AVG]: 0.587 vs. HC [AVG]: 0.615; $p\text{-value} = 0.085$). This result is expected, considering the complexity of the symptoms in PD patients resulting in a mismatch between the acoustic biomarkers and clinical assessment i.e., outlier values in the features computed for PD patients. The results also show that LLaMA performed consistently better across almost all metrics. In the case of the PD reports, there are no significant differences in the top 3 performing models (LLaMA [AVG]: 0.676 vs. DeepSeek [AVG]: 0.673 vs. Mistral [AVG]: 0.672; $p\text{-value} > 0.8$). In the case of HC reports, both LLaMA and Mistral performed similar across all metrics and significantly better than all other models in all metrics (T5/Flan-T5/GPT-2; $p\text{-values} < 0.001$; DeepSeek $p\text{-values} < 0.05$). Although DeepSeek (AVG: 0.704) performed behind LLaMA (AVG: 0.763) and Mistral (AVG: 0.762), it is still significantly better than T5, Flan-T5, and GPT-2 ($p\text{-values} < 0.001$).

Figure 5 shows the overall average performance (ROUGE, BLEU, and BERT scores) across LLMs. Overall, the top-3 performing models are LLaMA, Mistral, and DeepSeek, which demonstrated a significantly higher performance compared to T5, Flan-T5, and GPT-2 ($p\text{-values} < 0.001$). LLaMA was the best performing model, closely followed by Mistral with no significant difference between both models (LLaMA [AVG]: 0.720 vs. Mistral [AVG]: 0.717; $p\text{-value} = 0.821$), however, LLaMA and Mistral were significantly better than DeepSeek (LLaMA vs. DeepSeek; $p\text{-value} < 0.01$; Mistral vs. DeepSeek; $p\text{-value} < 0.05$). Overall, scaling the model size helps in the mFDA generation task; except for GPT-2, which showed almost no improvement when using bigger models, likely due to GPT-2 design focus on text generation, limiting its adaptability for structured tasks such as generating mFDA reports. For T5, the model plateau at the base model with no significant differences between the base and large models (T5-220M [AVG]: 0.597 vs. T5-770M [AVG]: 0.587; $p\text{-value} : 0.299$).

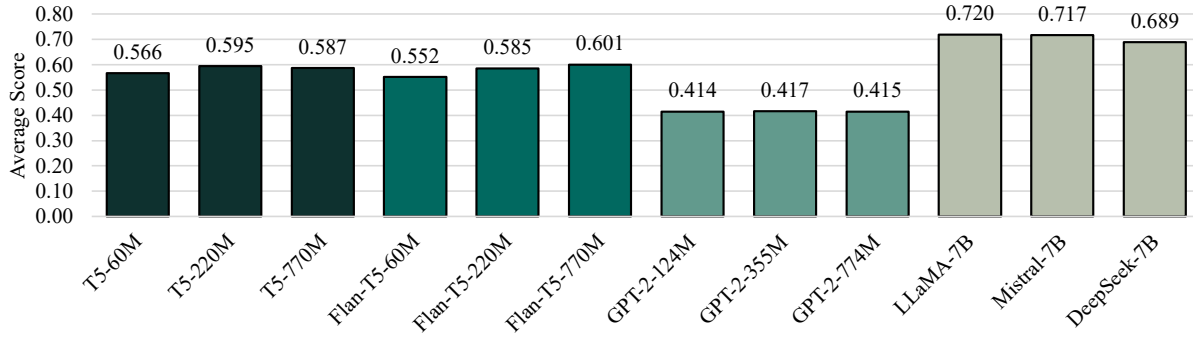


Figure 5. Overall average performance of the LLMs used in this work. Higher is better

Examples of LLaMA-generated PD and HC mFDA reports are shown in Figure 6. For reference, the input prompts corresponding to each case are also displayed in the figure. In both cases, LLaMA is able to create reports following the same structure and order of the target mFDA report i.e., category - rating - short description. As described in Section 2.1, the target ratings were performed by three phoniatricians using numerical values; thus, we calculate the median of the scores to get the target categorical rating of the subjects. In the case of the PD patient, the generated and real reports agreed on five out of seven categories. In the case of *Tongue*, the model rated the category as *Moderate*, which is one level lower than the target, which is *Severe*. In the case of *Larynx*, there is complete disagreement between the generated and target reports, i.e., there are at least two levels apart in the ratings *Mild* vs. *Severe*. Note that the ratings assigned by LLaMA agree with the thresholds reported in Table 3, except for *Larynx*, whose acoustic value goes beyond the *Severe*, but the model still rates this category as *Mild*. In the case of the HC report, all ratings assigned by LLaMA coincide with the ranges for the acoustic values in the input prompt (Table 3). The generated and target reports agree on five out of seven categories, with three interesting results. First, LLaMA rated *Lips* as *Mild* because its acoustic value is 0.01 points below the *Normal* threshold ($x \geq 1.62$, Table 3). Second, the category *Palate* was rated (by LLaMA) just one level below the target report (*Mild*). Third, LLaMA rates *Tongue* as *Severe*, which completely agrees with the target report. This is important because for a subject with mostly *Normal* and *Mild* ratings, the model still rates each category individually to produce the corresponding description based on the acoustic biomarker.

4 Conclusions

This paper presents a novel approach to generating mFDA-based clinical reports using LLMs and acoustic biomarkers, providing control over the input, thus improving interpretability and transparency. The results showed that LLMs effectively learn how to associate numerical values (acoustic features) to categorical ratings based on the mFDA scale and generate a structured report that considers different aspect of the speech production process. Large models (7 billion parameters) were able to

Input prompt (PD patient)	
Breathing: 71. Lips: 0.51. Palate: 0.44. Larynx: 1.47. Monotonicity: 0.041. Tongue: 0.028. Intelligibility: 0.66	
Generated (PD patient)	Ground truth (PD patient)
Breathing (Severe): control over respiration is grossly distorted. Lips (Severe): unable of maintaining good lip closure, resulting in poor production of sounds. Palate (Severe): speech quality is extremely deteriorated. Larynx (Mild): minor difficulties without noticeable speech problems. Monotonicity (Mild): minimal difficulties to change tone and variable pace. Tongue (Moderate): improper tongue control. Intelligibility (Mild): speech is mostly comprehensible.	Breathing (Severe): control over respiration is grossly distorted. Lips (Severe): unable of maintaining good lip closure, resulting in poor production of sounds. Palate (Severe): speech quality is extremely deteriorated. Larynx (Severe): unable to maintain a proper intonation and loudness variations during a conversation. Monotonicity (Mild): minimal difficulties to change tone and variable pace. Tongue (Severe): severe tongue control. Intelligibility (Mild): speech is mostly comprehensible
Input prompt (HC subject)	
Breathing: 34. Lips: 1.61. Palate: 0.98. Larynx: 0.21. Monotonicity: 0.054. Tongue: 0.035. Intelligibility: 0.89	
Generated (HC subject)	Ground truth (HC subject)
Breathing (Normal): normal breathing control. Lips (Mild): Inconsistent lip closure, but overall good control. Palate (Normal): normal production of velar sounds. Larynx (Normal): can maintain a normal phonation and produce clear speech. Monotonicity (Normal): person can produce variable tones and/or regular pace during speech. Tongue (Severe): severe tongue control. Intelligibility (Normal): speech is clear.	Breathing (Normal): normal breathing control. Lips (Normal): normal coordination of lips. Palate (Mild): mild abnormality. It does not affect speech production. Larynx (Normal): can maintain a normal phonation and produce clear speech. Monotonicity (Normal): person can produce variable tones and/or regular pace during speech. Tongue (Severe): severe tongue control. Intelligibility (Normal): speech is clear

Figure 6. Example of generated and mFDA reports from a PD patient (top) and a HC subject (bottom). The reports were generated with LLaMA-7B. The letters in *Green* indicate agreement between the LLM and the ground truth report, *Orange* indicate disagreement within one level (e.g. *Normal-Mild*), and *Red* means complete disagreement. Note that for the HC subject, the **Tongue** is properly rated as *Severe*, even when all other categories are rated *Normal* or *Mild*

generate the mFDA reports with higher quality than smaller models (<800 million parameters). These results are not surprising, however, we also observed that generation quality does not entirely depends on model scalability. In particular, we showed that LLaMA, Mistral, and DeepSeek-Qwen2.1 achieved similar results, however, the overall average performance of DeepSeek was significantly lower than LLaMA ($p\text{-value}<0.01$) and Mistral ($p\text{-value}<0.05$). In the case of smaller models, both T5 and Flan-T5 benefit from size scalability, contrary to GPT-2, which not only exhibit the worse performance but also showed no improvement when finetuning three size versions of this model. The results also showed that the best performing models (LLaMA, Mistral, and DeepSeek), struggled more to generate mFDA reports from PD patients compared to HC. These results are not surprising, considering that the complexity of the symptoms in PD patients result in outliers in the acoustic biomarkers (compared to HC subjects), leading to more mismatches between generated and ground truth reports. In addition, the HC reports contain more *Normal/Mild* categorical ratings, thus, the performance metrics are higher for the controls. However, generating structured mFDA reports is not a trivial task, as it was required to use models with 7B parameters to generate high quality reports, and even then, not all models performed the same. Regarding the training data, the simulated data enable the models to learn how to structure the reports by associating the acoustic biomarkers to the categorical ratings and corresponding descriptions; however, a limitation is that the numerical values used to simulate the prompts (and mFDA reports) constrain the model to assign categorical ratings within a range of values that might not account for outliers. Another consideration is the correlation between acoustic biomarkers and ground truth categorical ratings. PD/HC mFDA reports are created based on the average score given by three phoniatricians (inter-rater agreement of 0.87), which limits the correspondence between acoustic features and perceptual analysis. We showed that the selected biomarkers are significantly correlated with the categorical ratings, except for *Larynx* ($r_{pb} : 0.06$). Note that for this mFDA category we only focused on vowel phonation features, because that was the task used by the phoniatricians to perceptually evaluate this mFDA category, however, features extracted from different task might correlate better with this category.

Data Availability

Due to the confidential nature of speech data, we are unable to make our data publicly available. Access to the data can be made through reasonable requests to the GITA Lab and will be subject to local ethics clearances. Please email co-author rafael.orozco@udea.edu.co for requesting access to the data.

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Author contributions statement

T. Arias-Vergara contributed to the conceptualization, formal analysis, and investigation. T. Arias-Vergara and P. A. Pérez-Toro contributed to the methodology. T. Arias-Vergara, N. Vysotskaya, and L. Buess contributed to the writing-original draft preparation. T. Arias-Vergara, P. A. Pérez-Toro, S. T. Arasteh, J. R. Orozco-Arroyave, M. Schuster, E. Nöth, A. Maier, N. Vysotskaya, and L. Buess contributed to the writing-review and editing. T. Arias-Vergara, P. A. Pérez-Toro, and N. Vysotskaya contributed to the visualization.

Additional information

Competing interests

The authors declare no competing interests.